Biological Psychology 86 (2011) 337–348

Contents lists available at ScienceDirect

Biological Psychology

journal homepage: www.elsevier.com/locate/biopsycho

The neural correlates of cognitive effort in anxiety: Effects on processing efficiency

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ARTICLE INFO

Article history:
Received 4 September 2010
Accepted 24 December 2010
Available online 26 January 2011

Keywords:
Anxiety
Attentional control
ERP
CNV
Processing efficiency
Effort

ABSTRACT

We investigated the neural correlates of cognitive effort/pre-target preparation (Contingent Negative Variation activity; CNV) in anxiety using a mixed antisaccade task that manipulated the interval between offset of instructional cue and onset of target (CTI). According to attentional control theory (Eysenck et al., 2007) we predicted that anxiety should result in increased levels of compensatory effort, as indicated by greater frontal CNV, to maintain comparable levels of performance under competing task demands. Our results showed that anxiety resulted in faster antisaccade latencies during medium compared with short and long CTIs. Accordingly, high-anxious individuals compared with low-anxious individuals showed greater levels of CNV activity at frontal sites during medium CTI suggesting that they exerted greater cognitive effort and invested more attentional resources in preparation for the task goal. Our results are the first to demonstrate the neural correlates of processing efficiency and compensatory effort in anxiety and are discussed within the framework of attentional control theory.

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1. Introduction

Anxiety is a prominent and pervasive affect that occurs when our current or future goals are threatened (Eysenck, 1997). The attentional control theory of anxiety (ACT; Eysenck et al., 2007), was put forward to account for the detrimental effects of anxiety on cognitive performance. Central to the predictions of this theory was the assumption that anxiety affects the central executive functions of working memory, namely the inhibition and shifting functions, involved in successful task performance. Inhibition involves attentional control to resist interference from distracting and task-irrelevant stimuli, and shifting involves attentional control to maintain focus on changing task demands. ACT predicts that anxiety diminishes efficiency on tasks that involve central executive functions that depend on goal-directed attention allocation to be successful. By increasing the effects of stimulus driven processes over top down mechanisms anxiety is thought to reduce the influence of the goal-driven attentional system (Corbetta and Shulman, 2002) thus resulting in impaired cognitive task performance. There is accumulating evidence to support these predictions (see Derakshan and Eysenck, 2009, for a review).

ACT makes an important distinction between processing efficiency and performance effectiveness. Effectiveness refers to the quality of the performance outcome and is usually measured by behavioural indices such as response accuracy, while efficiency (normally measured by response latency) refers to the relationship between performance effectiveness and the manner in which resources are invested in task performance. It is argued that decreased efficiency is related to more resources invested to attain a given performance level. There is increasing evidence to support the prediction that anxiety impairs processing efficiency more than effectiveness (e.g. Ashcraft and Kirk, 2001; Elliman et al., 1997; Richards et al., 2000; Hadwin et al., 2005; Hayes et al., 2008, 2009; Johnson and Gronlund, 2009; Murray and Janelle, 2003, 2007; Visu-Petra et al., 2009; Sadeh and Bredemeier, 2010; see Derakshan and Eysenck, 2009, for a review). Within Eysenck et al. (2007) it is also predicted that anxious individuals make use of '*compensatory strategies*' such as enhanced effort or increased use of resources to prevent performance decrements and to maintain performance levels comparable to that of low-anxious individuals.

There is much behavioural research that supports the assumption that anxiety impairs processing efficiency however the majority of this research has used indirect behavioural measures to assess performance efficiency and very few have assessed the anxiety–performance relationship with specific focus on the shifting and inhibition functions. The goal of the current study is to evaluate assumptions of the ACT using direct measures of processing efficiency and cognitive effort.

Most behavioural studies have quantified performance effectiveness in terms of error rates but assessment of processing efficiency has been more indirect. Time taken to respond correctly has been used as one indicator of processing efficiency; thus the

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doi:10.1016/j.biopsycho.2010.12.013

longer it takes to achieve a task goal, the lower the processing efficiency (Eysenck et al., 2007). Richards et al. (2000) compared high- and low-anxious individuals' performance on a memory task in which a string of numbers had to be maintained in working memory during an inferential reasoning task. They found no group differences in accuracy of recalled digits. However, high-anxious individuals spent longer studying the memory item when it was presented at the beginning of each trial, which the authors argue is suggestive of a motivation to perform correctly.

Processing efficiency has also been assessed as a function of the *amount of effort* exerted to achieve performance effectiveness. According to ACT, high-anxious individuals exert more effort (and hence utilise more cognitive resources) than the low-anxious but achieve only comparable effectiveness. Hadwin et al. (2005) required children to perform forward and backward digit span tasks that tapped the phonological and central executive components of working memory. Reaction time on correct trials and scores on a self-report effort scale were used to measure performance efficiency while error rate was taken as a measure of effectiveness. Hadwin et al. (2005) found no anxiety-related differences in accuracy, however, high- compared with low-anxious children were significantly slower on the backward digit span task and reported increased mental effort on the forward digit span task supporting predictions of ACT. Similar results were reported by Smith et al. (2001) who examined the effect of anxiety on performance in a team sport setting. These findings are in line with the notion that increased anxiety could motivate increased effort in an attempt to reduce processing inefficiency to achieve the task goal.

Despite their significant value, response time and self-report measures are considered to be indirect measures of processing efficiency (or amount of processing resources used) as they reflect the *outcome* of processing. A more direct method, which has gained popularity in anxiety research with the advent of neuroimaging techniques, is to assess brain activity during task performance. Fales et al. (2008) assessed anxiety modulation of cognitive control mechanisms using fMRI while high- and low-anxious individuals performed a demanding working memory (verbal n-back) task. It was argued that cognitive control could either be exerted in a sustained or a transient manner depending on the task requirements. In sustained (or '*proactive*') control, task-goals are kept active throughout intervals that require high levels of control. This is believed to allow for more top–down control over processing as it promotes early preparation (or biasing) of attentional resources. In transient (or '*reactive*') control, task-goals might not be kept active after the initial encoding but are reactivated *only when* required. This type of control is thought to be useful in situations when changes in task demand are unpredictable. Fales et al. (2008) found that high-versus low-anxious participants exhibited *reduced sustained* activation but *increased transient* activation in prefrontal brain areas involved in executive control (namely dorsolateral and ventrolateral PFC). Notably there were no anxiety-related differences in behavioural performance. Fales et al. (2008) conclude that enhanced anxiety impairs "optimal" activation of prefrontal control mechanisms (i.e. sustained control) but that this is compensated for by the increased transient/reactive control. These findings are in line with assumptions of ACT that anxious individuals use compensatory strategies but achieve outcomes that are only comparable to the low-anxious.

1.1. Current experiment

The current experiment sought to directly evaluate the prediction that anxiety impairs processing efficiency in contexts where the shifting function of the central executive is required (Eysenck et al., 2007). We measured well-established neural correlates of cognitive effort and resource allocation, namely the Contingent Negative Variation, (CNV; Walter et al., 1964) during performance on a modified mixed antisaccade task. In the standard antisaccade task (Hallet, 1978) participants must suppress the natural tendency to saccade (a fast eye-movement) toward an abrupt peripheral stimulus and instead to look *away* to it's opposite direction. Antisaccade latencies and error rates are compared to that of 'prosaccade' trials where the abrupt peripheral stimulus must be fixated as soon as it appears. Recently, a number of studies have tested the predictions of ACT using the antisaccade task. For example, Ansari et al. (2008) examined how anxiety impaired the shifting function. In a similar fashion, the effects of anxiety on the inhibition function were examined by Derakshan et al. (2009) as well as Ansari and Derakshan (2010). The results of these studies collectively indicated that anxiety impaired the inhibition and shifting functions of working memory by resulting in longer antisaccade latencies. Other work using the antisaccade task has also found evidence for impaired inhibitory processes in response to facial expressions of emotion in social anxiety (Wieser et al., 2009).

In the current experiment anti- and prosaccade tasks were presented in a randomly interleaved order and the required saccade type (anti- or prosaccade) was instructed at the beginning of each trial using a coloured cue. Hence on some trials ('switch' trials) participants had to shift between task-sets depending on the colour of the instructional cue (see Section 2). Many studies have used this task to assess shifting ability in clinical non-clinical contexts (e.g. Ansari et al., 2008; Barton et al., 2002; Cameron et al., 2009; Cherkasova et al., 2002; Dyckman et al., 2007; Ethridge et al., 2009; Johnston et al., 2007; Mueller et al., 2007). The current study uniquely varied the interval between instructional cue and target (cue-target interval; CTI), a key parameter in the mixed antisaccade task, to compare anticipatory/preparatory CNV activity of low- and high-anxious individuals. This design was based on research suggesting that the amplitude of CNV activity during cue-target intervals provides a direct indication of the amount of processing resources allocated (or effort exerted) to prepare brain areas involved in responding (e.g., Bender et al., 2004; Cui et al., 2000; Jennings and van der Molen, 2005; Rösler et al., 1997).

1.2. ERP correlates of cognitive effort and processing efficiency

Allocation of cognitive resources in preparation for an imperative action has routinely been assessed by examination of slow-wave cortical potentials (SCP; see Rösler et al., 1997; Kok, 2001; Kotchoubey, 2006; Jennings and van der Molen, 2005). It has been suggested that negative SCP shifts reflect *preparation* of cortical resources for an anticipated activity while positive SCP shifts are related to the *consumption* of these resources in actual activity (see Kotchoubey, 2006). Among the varieties of SCPs, the Contingent Negative Variation (CNV; Walter et al., 1964), which is of central focus in the current study, has received a great deal of interest.

A CNV is a slow negative deflection produced when a warning signal (a cue) induces preparation for an imperative event and it is believed to reflect activation (or preparation) of cortical areas that are required to respond to or process the expected stimulus or event. This potential is maximal over frontal scalp regions after presentation of the warning cue (e.g. Gómez et al., 2007; Rosahl and Knight, 1995). Amplitude of a CNV shift has been associated with processes relating to warning cue evaluation, estimation of target onset time and cognitive preparation of cortical networks involved in the pending response (e.g. Bender et al., 2004; Cui et al., 2000; Ikeda et al., 1996; Rösler et al., 1997; Yazawa et al., 1997). This amplitude is widely believed to be influenced by the degree of cognitive effort due to motivation and/or task complexity. Rösler et al. (1997), in their comprehensive account, suggest that SCP (including CNV) amplitudes reflect the extent/amount of resources allocated to a cognitive process while its scalp topogra-

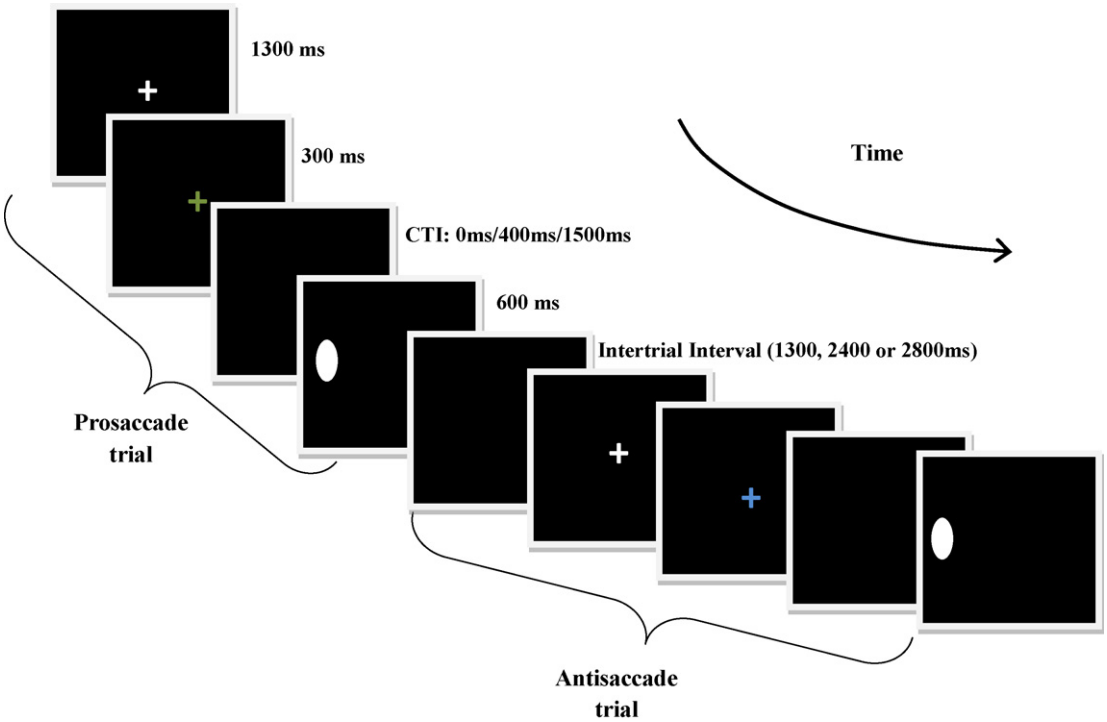

Fig. 1. A schematic example of a 'switch' antisaccade trial in which the colour of the cross indicates the required eye-movement (anti- or prosaccade).

phy reflects the task-specific process that the resources are "fed in to". Rösler et al. (1997) also suggest that the times-course of SCP shifts (i.e. their continuation or dissipation over time) is related to the duration of the cognitive process; the pattern changing during the course of the cognitive task depending on the requirements. Further to this, Jennings and van der Molen (2005) suggest that maintaining cognitive preparation is effortful and hence 'optimal' preparedness is time-limited (the length of which will vary with task characteristics). This view also agrees with findings of the task-switching literature that typically shows brief preparatory periods to be more effective in reducing switch costs (e.g. Mayr and Kliegl, 2003).

1.3. Predictions

Increasing preparation time in anticipation of an imperative event has consistently been found to reduce reaction times (e.g. Jennings and van der Molen, 2005; Meiran, 1996). If anxiety motivates greater effort and resource allocation to task performance as ACT suggests, then two predictions can be formulated. First, we would expect high-anxious individuals to be faster (resulting in switching performance that is comparable to low-anxious individuals) when there is time to prepare the correct response versus no preparation. Second if this improvement is due to effortful processing and increased resource allocation, then the effects would be evident in greater frontal CNV amplitudes in the high-anxious group.

2. Method

2.1. Participants

Thirty-two participants (mean age = 24.42 ± 5.06, min = 18, max = 37) were selected from a larger University of London subject pool based on their scores on the State-Trait Anxiety Inventory (STAI; Spielberger et al., 1983). High-anxious (HA) participants were classified as those who scored 44 and above on the trait anxiety scale of the STAI while those who scored 34 and below on this scale were classified as low-anxious (LA). All participants had normal or corrected-to-normal vision. Each individual was tested in a separate laboratory session lasting approximately 3 h and was reimbursed (£24) for their contribution.

2.2. Experimental task and procedure

Anti- and prosaccade tasks were pseudo-randomly interleaved resulting in 'repeat' and 'switch' Trial types. 'Repeat' trials were those in which the instructed eye-movement (anti- or prosaccade, indicated at the beginning of every trial) was the same as that on the trial immediately preceding it (e.g. an antisaccade preceded by an antisaccade). 'Switch' trials were those in which the instructed eye-movement was different to that on the preceding trial (e.g. antisaccade preceded by a prosaccade).

As illustrated in Fig. 1, each trial began with a central fixation cross. After 1300 ms the cross changed colour to either green or blue indicating the required saccade on the current trial (e.g. green indicating anti- and blue indicating prosaccade). Offset of the instructional cue after 300 ms was followed by the target ($1^\circ \times 3^\circ$) appearing 11° to either the left or right side of the screen (duration: 600 ms). Target appearance to the left and right side of the screen was randomly and equally distributed across trials. On one-third of trials, target onset followed cue offset immediately (short cue-target interval trials, CTI). On another third of the trials, target appeared after a 400 ms lag (medium CTI) during which everything was extinguished from the screen. On the final third of trials target appeared after a 1500 ms lag (long CTI trials). Participants were instructed to fixate the central cross and to focus on its colour until it disappeared and then to direct their gaze accordingly – either '*away*' from the target (on antisaccade trials) to the opposite side of the screen, or '*toward*' it (on prosaccade trials), as soon as it appeared. Colour–saccade type associations were counterbalanced in a within-subject design. Importance of speed and accuracy of eye-movements were emphasised.

Each experimental session began with participants completing the consent form and the STAI measure of trait anxiety. After a brief oral demonstration of the equipment, the electrode cap was applied and electrode impedances were reduced to ≤5 kΩ. Participants were informed of the task requirements and completed 24 practice trials at the beginning of the experiment. They were explicitly encouraged to maintain central fixation whenever the cross was on the screen and to minimise movements and blinks during the experimental trials. They completed 12 blocks of 102 trials each block separated by a few minutes of rest. Participants were reminded of the colour–saccade association half-way through each block. In total there were 400 short, 400 medium and 400 long CTI trials (half anti- and half prosaccade trials) which were presented randomly making it difficult for participants to predict target onset. There were equal numbers of 'switch' and 'repeat' Trial types which were pseudo-randomly presented (randomisation based on previous research; Kieffaber and Hetrick, 2005). In line with task-switching literature, the first experimental trial in each block was excluded from analysis. This design resulted in 1224 trials in total (including 24 first trials).

2.3. Data acquisition

Using BrainVision Recorder (BrainAmps), we recorded EEG data from 30 Ag/AgCl electrodes (Fpz, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP6, P7, P3, Pz, P4, P8, PO7, O1, O2, PO8) mounted in an elastic electrode cap according to the international 10%-system. All electrodes were referenced to the left mastoid (M1) and later re-referenced to the mean of both mastoids. Forehead was used as ground. Horizontal eye-movements (HEOG) were recorded with electrodes placed laterally on the outer canthi of the eyes, and vertical eye-movements (VEOG) were recorded from an electrode placed below the right eye, (VEOG and FPz were used to detect blinks). Data was amplified between 0.1 and 125 Hz, sampled at 1000 Hz, and offline filtered with a bandpass frequency of 0.01–30 Hz.

2.4. Saccade analysis

Saccade onset was identified in the EOG as a rapid shift away from baseline (±50 μV). Matlab scripts were used to detect saccades (which were also inspected visually). Leftward and rightward direction of saccades were derived by calculating the difference between the potentials of the left and right HEOG electrodes while latency of saccade onset was defined as the point at which this difference exceeded 50 μV. Latency and directional accuracy of the first saccade after target onset were measured to evaluate performance.

An *incorrect* saccade was defined as the first saccade after target onset with amplitude greater than 50 μV *toward* the position of the peripheral target on antisaccade trials, or *away* from it on prosaccade trials. Latency of the first correct saccade was defined as the elapsed time between onset of the target and the beginning of a saccade toward the correct area of interest (i.e. the same side as the oval object on prosaccade trials, and the side mirroring the oval object on antisaccade trials). In line with common definitions (Fischer et al., 1993), trials in which saccade latency was less than 80 ms were excluded as anticipatory. Trials with saccade latencies exceeding 500 ms were also excluded from further analysis as were trials with other artifacts (blinks, large amplitude shifts before target onset, and eye-movements *before* target onset). Participant data was excluded from analysis if 40% or more trials were lost due to the above criteria.

2.5. Event-related potential analysis

We were interested in investigating how anxiety modulated CNV activity during the cue-target interval. EEG data was offline epoched into 700 ms (for short CTI), 1100 ms (for medium CTI) and 2200 ms (for long CTI) windows starting 100 ms prior to instructional cue presentation. The baseline was defined by the mean voltage over the initial 100 ms of the averaged epochs. In line with other antisaccade research (e.g. Everling et al., 1998a,b, 2001; Evdokimidis et al., 1997) contralateral, ipsilateral and midline responses in terms of recording sites with respect to the target position were obtained by pooling data from stimulus positions to the left and to the right. Grand average waveforms were plotted (see Section 3) showing cortical activity 100 ms before instructional cue onset to 100 ms after target onset for 'switch' and 'repeat' anti- and prosaccade trials.

We examined activity during the instructional cue by averaging across instruction presentation intervals of short, medium and long CTI trials. This analysis is not relevant to the aims of the current experiment and hence will not be presented here. To compare CNV activity between low- and high-anxious individuals we examined activity locked to cue offset during the medium and long CTIs separately. Activity during medium CTI trials was assessed in two 200 ms intervals ('early' and 'late') and during long CTI trials it was examined in two 200 ms intervals ('early', and 'intermediate') and one 1100 ms interval ('late'). We predicted greater frontal CNV activity in high-anxious compared with low-anxious individuals during medium and long CTIs. Matlab was used to process and prepare data sets for analysis while SPSS statistical package was used for data analysis.

2.6. Statistical analysis

2.6.1. Correct saccade latencies and percentage saccade errors

Mean latencies of valid *correct* saccades and percentage of *incorrect* saccades were analysed using two mixed ANOVAs with Group (low-anxious; LA, high-anxious; HA) as the between-subject factor and Task (anti-, prosaccade), Trial type (switch, repeat) and CTI (short, medium, long) as the within-subject factors. Greenhouse–Geisser corrections were applied if assumptions of sphericity were violated. Descriptive statistics are expressed as means ± standard deviation from the mean.

2.6.2. Cortical potentials prior to correct saccades

Statistical comparisons of ERP activity were conducted on data from nine electrodes (F3, F4, Fz, C3, C4, Cz, P3, P4, Pz). Mixed ANOVAs examined cortical potentials on correct anti- and prosaccade trials. In these analyses, Task (anti-, prosaccade), Trial type (switch, repeat), and Site (F3/F4, Fz, C3/C4, Cz, P3/P4, Pz) were the within-subject factors while Group (low-anxious; LA, high-anxious; HA) was the between-subject factor.[1] Greenhouse–Geisser corrections were applied if assumptions of sphericity were violated. Simple effects analyses were carried out where appropriate. Data are expressed as the mean ± standard deviation from the mean.

[1] The factor Laterality (Contralateral and Ipsilateral to target) did not yield any significant interactions with the main variables of interest and therefore following the suggestion of two anonymous reviewers the analyses collapses across laterality.

3. Results

3.1. Participants

Data for 24 participants were analysed,[2] half were low-anxious (LA; mean trait anxiety score = 28.08 ± 4.9, min = 24, max = 34) and half high-anxious (HA; mean trait anxiety score = 49.33 ± 4.1, min = 44, max = 56).

3.2. Mean correct saccade latencies

A 2 × 2 × 2 × 3 mixed ANOVA on correct saccade latencies with Group (LA, HA) as between- and Task (anti-, prosaccade), Trial type (switch, repeat) and CTI (short, medium, long) as within-subject factors revealed a number of significant main effects and interactions. There was a main effect of Task, $F(1,22) = 176.67$, $p < .001$, reflecting slower antisaccade latencies ($M = 200.84 \pm 19.27$ ms) compared with prosaccade ($M = 156.31 \pm 21.18$ ms), as well as a main effect of Trial type, $F(1,22) = 124.97$, $p < .001$, reflecting slower saccades on 'switch' trials ($M = 183.32 \pm 18.47$ ms) compared with 'repeat' ($M = 173.82 \pm 18.68$ ms), and a main effect of CTI, $F(2,44) = 116.57$, $p < .001$, to show that saccades were slower on short CTI trials ($M = 203.79 \pm 23.93$ ms) compared with medium ($M = 163.83 \pm 17.92$ ms; $t(23) = 11.49$, $p < .001$) and long ($M = 168.1 \pm 19.02$ ms; $t(23) = 10.49$, $p < .001$). There were also two-way interactions of Task × Trial type, $F(1,22) = 20.52$, $p < .001$, Task × CTI, $F(2,44) = 10.78$, $p < .001$; and Trial type × CTI, $F(2,44) = 19.87$, $p < .001$, as well as a three-way interaction of Task × Trial type × CTI, $F(2,44) = 12.45$, $p < .001$. Analyses decomposing these interactions with anxiety group are discussed below.

3.3. Anxiety-related group differences

Of major importance were the interactive effects with anxiety group. There was a significant CTI × Group interaction, $F(2,44) = 4.99$, $p < .02$, suggesting that increasing preparation time had differential effects on overall performance of HA and LA individuals, such that correct saccade latencies improved significantly in both groups when CTI was increased from 0 ms (short) to 400 ms (medium) (LA: mean difference = 30.89 ms, $t(11) = 6.93$, $p < .001$; HA: mean difference = 49 ms, $t(11) = 12.35$, $p < .001$). Improvement was significantly greater in the HA compared with the LA ($t(22) = 3.04$, $p = .006$) (see Fig. 2). Latencies were also faster when CTI was increased from 0 ms (short) to 1500 ms (long) (LA: mean difference = 31.7 ms, $t(11) = 6.54$, $p < .001$; HA: mean difference = 39 ms, $t(11) = 8.45$, $p < .001$). However, increasing preparation time from medium to long did not result in significant improvement in either group, it had a negative effect on HA (medium: $M = 155.8 \pm 13.25$ ms, long: $M = 165.15 \pm 18.09$ ms, $t(11) = 2.84$, $p < .01$) and the LA individuals did not show any improvement (medium: $M = 171.85 \pm 18.85$ ms, long: $M = 171.05 \pm 20.26$ ms, $t < 1$). Thus, while preparation time facilitated correct saccade latencies to an extent, increasing this beyond a certain level did not add to the performance of both groups. There was some indication that on medium CTI trials HA participants were faster to generate correct saccades ($M = 151.8 \pm 13.25$ ms) compared with the LA ($M = 171.85 \pm 18.85$ ms), $t(22) = 2.41$, $p = .03$.

There was also a significant four-way interaction of Task × Trial type × CTI × Group, $F(2,44) = 4.34$, $p < .03$, to consider the effect of CTI *on switching* performance in LA and HA groups (descriptive statistics in Table 1). This interaction revealed that on short CTI trials, HA participants were slower on 'switch' compared with 'repeat' antisaccade trials, $t(11) = 4.84$, $p = .001$, but this difference missed significance in the LA group, $t(11) = 2.11$, $p = .06$. The LA and HA groups differed from each other on latency 'switch cost' (i.e. the latency difference between 'switch' and 'repeat' trials), $t(22) = 2.55$, $p = .02$ (see Table 1 for descriptives). However, both groups showed comparable switch costs in antisaccade latency on medium CTI trials $t(23) < 1$. Long CTIs on the other hand, resulted in slower 'switch' versus 'repeat' antisaccade latencies in both HA ($t(11) = 2.36$, $p = .04$) and LA ($t(11) = 4.2$, $p = .001$) individuals. Overall, these results suggest that when there was very little time to prepare before target onset (short CTI trial), HA individuals were slower to generate antisaccades in the 'switch' compared with the 'repeat' condition whereas LA individuals were unaffected by the requirement to switch on these trials. When there was some preparation time prior to target onset (medium CTI trials) latency switch costs were eliminated in HA and LA individuals. However, extending preparation time further (long CTI) resulted in antisaccade latency switch costs in both groups.

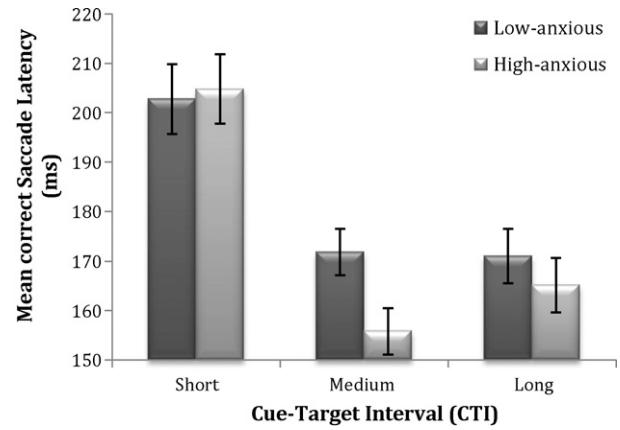

Fig. 2. Mean correct saccade latencies (with SE) after short (0 ms), medium (400 ms) and long (1500 ms) cue-target intervals.

On prosaccade trials, HA and LA individuals exhibited comparable latency switch costs on short (LA: $t(11) = 5.42$, $p < .001$; HA $t(11) = 8.49$, $p < .001$), long (LA: $t(11) = 3.24$, $p < .01$; HA: $t(11) = 3.89$, $p < .01$) and medium (LA: $t(11) = 6.84$, $p < .001$; HA: $t(11) = 2.57$, $p = .03$) CTI trials (see Table 1 for descriptive statistics).

3.4. Percentage saccade errors

Fig. 3 depicts percentage anti- and prosaccade errors on short, medium and long CTIs trials. Participants made more errors on antisaccade trials ($M = 11.11 \pm 10.46\%$) compared with prosaccade trials ($M = 3.09 \pm 3.11\%$), $t(23) = 3.78$, $p < .001$. There were also more errors on 'switch' ($M = 10.04 \pm 6.75\%$) versus 'repeat' ($M = 4.15 \pm 5.06\%$) trials, $t(23) = 8.14$, $p < .001$, and on short ($M = 11.52 \pm 7.69\%$) compared with medium ($M = 5.81 \pm 6.18\%$), $t(23) = 6.33$, $p < .001$, and long ($M = 3.96 \pm 4.31\%$), $t(23) = 6.76$, $p < .001$, CTI trials. This observation was supported by a 2 × 2 × 3 × 2 mixed ANOVA on percentage error with Group (LA, HA) as the between- and Task (anti-, prosaccade), Trial type (switch, repeat) and CTI (short, medium, long) as the within-subject factors which revealed significant main effects of Task, $F(1,22) = 31.67$, $p = .001$, Trial type, $F(1,22) = 63.92$, $p < .001$, and CTI, $F(2,44) = 39.52$, $p < .001$. The analysis also yielded significant Task × Trial type, $F(1,22) = 13.73$, $p < .001$, Task × CTI, $F(2,44) = 13.42$, $p < .001$, Trial type × CTI, $F(2,44) = 54.32$, $p < .001$, and Task × Trial type × CTI, $F(2,44) = 12.06$, $p < .001$, interactions. Importantly for the purposes of the current investigation, this

[2] Data for eight individuals were excluded from analysis as too many trials were excluded based on the exclusion criteria (see Section 2). The final data set lost 8.87% and 8.01% antisaccade trials and 6.91% and 7.05% prosaccade trials in (LA and HA groups respectively). There were no group differences in lost data.

Table 1
(a) Descriptive statistics for mean correct antisaccade latencies (with std. dev.).

	Short CTI			Medium CTI			Long CTI		
	Repeat	Switch	Paired sample $t(11)$=	Repeat	Switch	Paired sample $t(11)$=	Repeat	Switch	Paired sample $t(11)$=
Low-anxious	217.2 (26.22)	221.32(27.71)	2.11, $p = .06$	193.02(21.61)	194.22 (20.4)	N.S.	187.84 (21.0)	198.84(20.02)	4.2, $p < .001$
High-anxious	218.44(25.78)	232.23(22.94)	4.84, $p = .001$	179.55(15.75)	183.26(17.12)	N.S.	189.31(21.71)	194.82(22.46)	2.36, $p = .04$
Ind. sample $t(23)$=	N.S.	N.S.	–	N.S.	N.S.	–	N.S.	N.S.	–

(b) Descriptive statistics for mean correct prosaccade latencies (with std. dev.)

	Short			Medium			Long		
	Repeat	Switch	Paired sample $t(11)$=	Repeat	Switch	Paired sample $t(11)$=	Repeat	Switch	Paired sample $t(11)$=
Low-anxious	170.99(28.79)	198.87(30.14)	5.42, $p < .001$	145.75(20.58)	154.43(20.99)	6.84, $p < .001$	144.92(24.76)	152.61(23.89)	3.24, $p < .01$
High-anxious	173.33 (27.4)	195.3 (28.17)	8.49, $p < .001$	128.36(16.28)	132.04(16.69)	2.57, $p = .03$	134.53(19.51)	141.94 (20.4)	3.89, $p < .01$
Ind. sample $t(23)$=	N.S.	N.S.	–	2.3, $p = .03$	2.89, $p < .01$	–	N.S.	N.S.	–

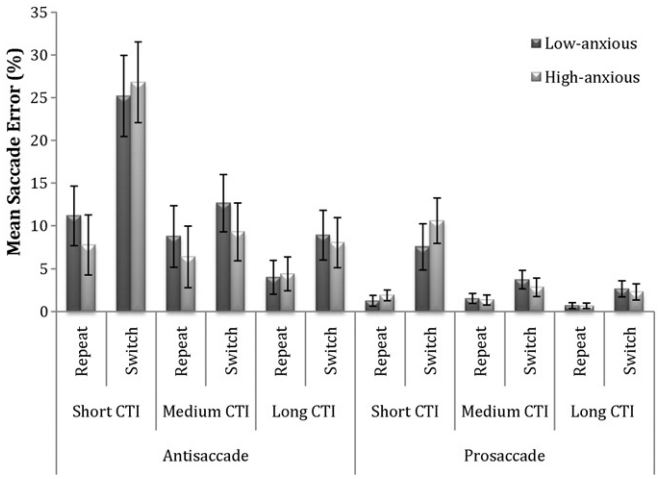

Fig. 3. Percentage saccade error (with SE) on anti- and prosaccade trial for each group.

analysis revealed no significant anxiety group-related differences in percentage error rates, $Fs < 1$. This result replicates our previous findings (Ansari and Derakshan, 2010; Ansari et al., 2008; Derakshan et al., 2009).

3.5. Event-related potentials

3.5.1. Activity during medium CTI: overview

Fig. 4 depicts grand average waveforms across the scalp during the medium CTI period on correct antisaccade trials for low-anxious and high-anxious individuals. Grand averages show a sharp negative deflection at the offset of the instructional cue at frontal electrode sites. The negativity appeared to be greater in HA compared with LA individuals and this difference persisted after target onset. Activity at parietal sites remained positive after cue offset. There was a momentary negative dip at ~150 ms which was then followed by a positive deflection which persisted until target onset. HA and LA did not exhibit differential activity at parietal sites. As depicted by the grand averages in Fig. 5, the pattern of activity on correct prosaccade trials mirrored that on antisaccade trials. The observations in medium CTI trials were assessed at two intervals: the 'early' (0–200 ms after instructional cue offset) and 'late' (200–400 ms after instructional cue offset).

3.5.2. Early preparation interval (medium CTI): 0–200 ms after instruction offset

A $2 \times 2 \times 2 \times 6$ mixed ANOVA on cortical activity with Group (LA, HA) as the between- and Task (anti, prosaccade), Trial type (switch, repeat), and Site (F3/F4, Fz, C3/C4, Cz, P3/P4, Pz) as the within-subject factors yielded a significant main effect of Site, $F(5,110) = 24.21$, $p < .001$, and two interactions: Task × Site, $F(5,110) = 4.36$, $p = .03$, and Trial type × Site × Group, $F(5,110) = 6.70$, $p = .003$. The main effect of Site suggests that after instruction cue offset, activity at frontal and central sites was negative (maximal at Cz: −1.68 μV) compared to parietal sites (maximal at Pz: 2.26 μV). The Task × Site interaction revealed that in the first 200 ms after cue offset, cortical activity at central and parietal sites was greater on pro- compared to antisaccade trials (P3/P4 anti: 1.62 μV, pro: 2.2 μV, $t(23) = 2.28$, $p = .03$, and C3/C4 anti: −.2 μV, pro: .38 μV, $t(23) = 2.04$, $p = .05$).

Simple effects analyses decomposing the three-way Trial type × Site × Group interaction suggest that on prosaccade trials HA compared to LA individuals exhibited greater negativity on 'switch' trials at frontal (F3/F4 HA: −1.35 μV, LA: 1.0 μV, $t(22) = 2.37$, $p = .03$) sites. However, there were no group differences on 'repeat' prosac-

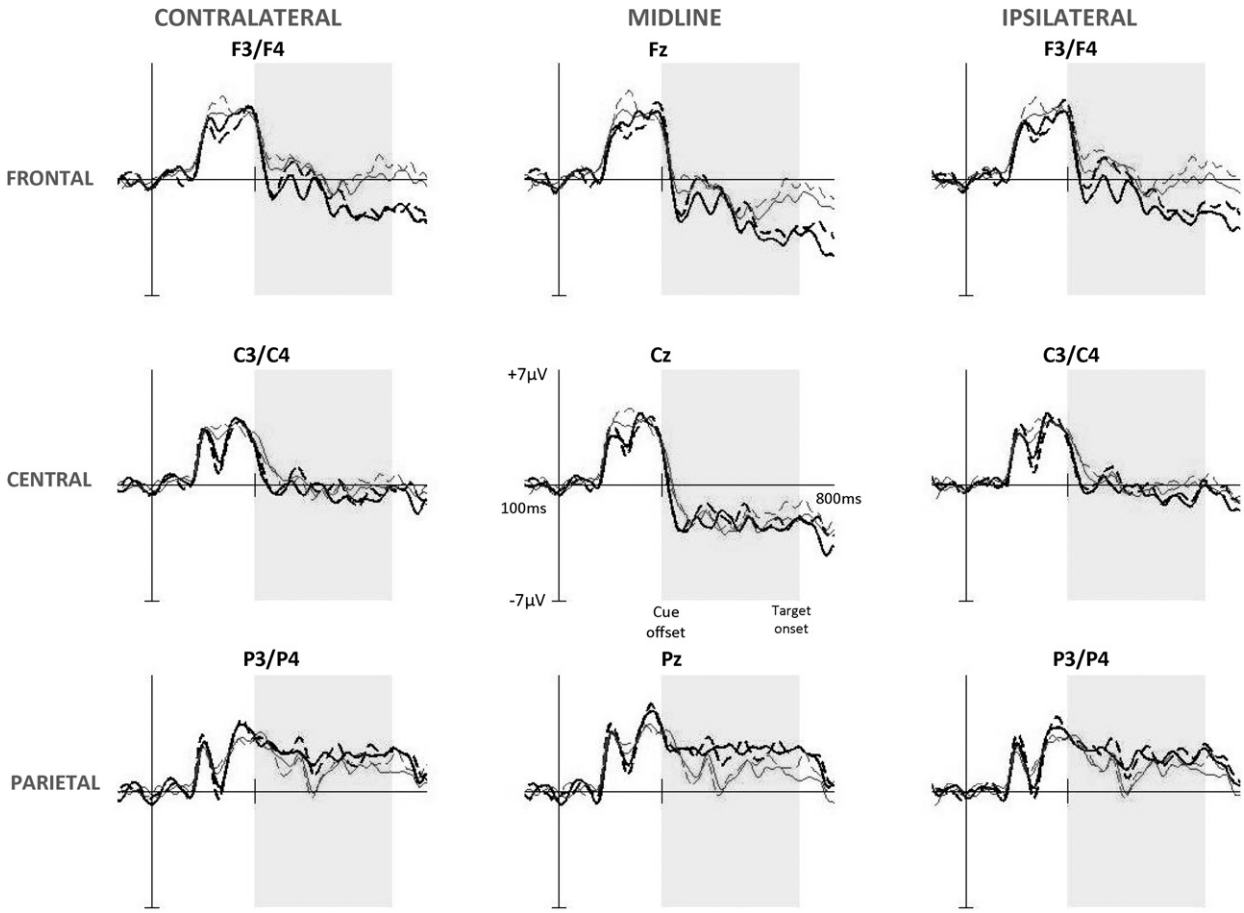

Fig. 4. Grand averages during medium CTI at contralateral, ipsilateral and midline electrodes for HA (dark black lines) and LA (grey lines) individuals on correct switch (solid lines) and repeat (dashed lines) antisaccade trials.

cade trials; *t*-tests N.S. Furthermore, HA individuals exhibited differential cortical activity between 'switch' and 'repeat' prosaccade trials at P3/P4 (switch: 2.47 μV, repeat: 1.82 μV, $t(11) = 2.61$, $p = .02$) whereas there was no differential activity between 'switch' and 'repeat' prosaccade trials in LA individuals. Anxiety did not modulate antisaccade performance in this period, $F < 1$.

3.5.3. Late preparation interval (medium CTI): 200–400 ms after instruction offset

A $2 \times 2 \times 2 \times 6$ mixed ANOVA on cortical activity with Group (LA, HA) as the between- and Task (anti, prosaccade), Trial type (switch, repeat), and Site (F3/F4, Fz, C3/C4, Cz, P3/P4, Pz) as the within-subject factors yielded a significant main effect of Site, $F(5,110) = 22.04$, $p < .001$, suggesting that activity at frontal sites was negative (maximal at Fz: −1.92 μV) and more positive at parietal sites (maximal at Pz: 1.7 μV). Central to the current study, the Site × Group interaction, $F(5,110) = 4.03$, $p = .02$, showed that HA compared to LA exhibited greater negativity at frontal sites (F3/F4 HA: −2.03 μV, LA: .61 μV, $t(22) = 2.83$, $p = .01$, and Fz, HA: −3.32 μV, LA: −.52 μV, $t(22) = 2.78$, $p = .01$). However, the groups exhibited comparable positivity at parietal sites (P3/P4 HA: 1.63 μV, LA: 1.5 μV, and Pz HA: 1.71 μV, LA: 1.74 μV, $ts < 1$).

3.5.4. Summary (medium CTI trials)

Offset of the instructional cue signalled the imminence of target onset and the requirement to make a speeded response. Overall, ERP activity during medium CTI trials shows that the response to this warning signal was a frontal CNV approximately 200 ms after cue offset that was greater in HA compared with LA individuals. In line with previous research, the frontal CNV continued until target onset reflecting a sustained state of 'preparedness' (Jennings and van der Molen, 2005). Anxiety did not modulate parietal activity.

3.5.5. Activity during long CTI: overview

Fig. 6 depicts grand average waveforms during the long CTI period on correct antisaccade trials for low-anxious and high-anxious individuals. Similar to medium CTI trials, these grand averages show that the offset of instructional cue induced a sharp negative deflection at frontal electrode sites which was greater for HA compared with LA individuals. This greater negativity in HA individuals began to dissipate toward baseline at ∼500 ms after cue offset such that when the target was presented, frontal activity in HA was comparable to that in LA individuals. Cue offset was followed by parietal positivity which continued for ∼150 ms before exhibiting a sharp negative dip followed by a positive deflection which persisted until ∼300 ms post cue offset. This positivity gradually dissipated toward baseline and was negative at target onset. Anxiety did not modulate activity at these parietal sites. As depicted by grand averages in Fig. 7, the pattern of activity on correct prosaccade trials was similar to that on antisaccade trials. These observations were examined in three separate intervals: 'early' (0–200 ms after cue offset) and 'intermediate' (200–400 ms after cue offset) and 'late' (400–1100 ms after cue offset).

3.5.6. Early preparation interval (long CTI): 0–200 ms after instruction offset

A $2 \times 2 \times 2 \times 6$ mixed ANOVA on cortical activity with Group (LA, HA) as the between- and Task (anti, prosaccade), Trial type

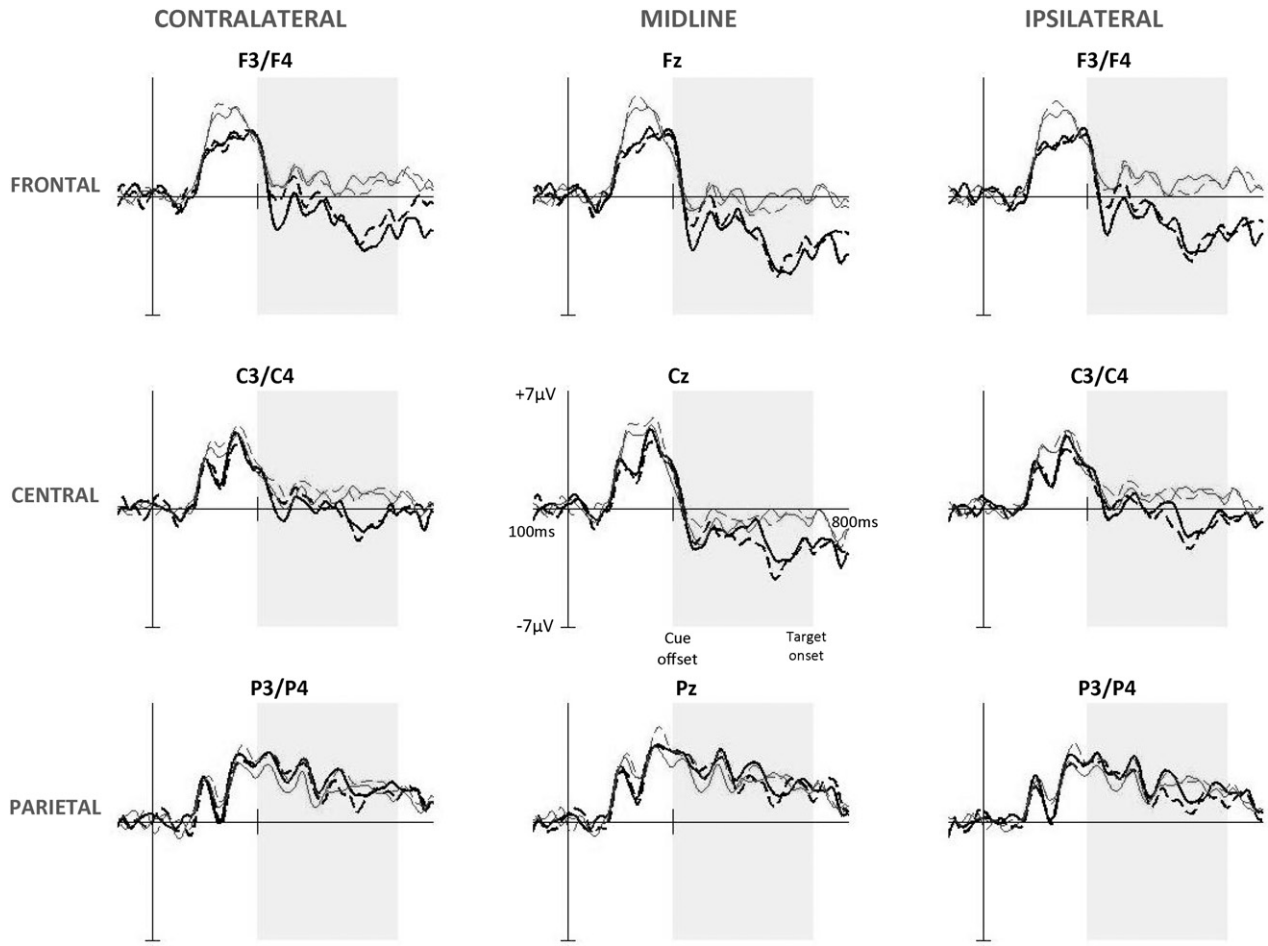

Fig. 5. Grand averages during medium CTI at contralateral, ipsilateral and midline electrode for HA (dark black lines) and LA (grey lines) individuals on correct switch (solid lines) and repeat (dashed lines) prosaccade trials.

(switch, repeat), and Site (F3/F4, Fz, C3/C4, Cz, P3/P4, Pz) as the within-subject factors yielded a significant main effect of Site, $F(5,110) = 24.64$, $p < .001$, a and Task × Trial type × Group, $F(5,110) = 5.22$, $p = .03$, interaction. The main effect of Site suggested that activity was more positive at parietal sites (maximal at Pz: 2.29 μV) whereas at frontal and central sites activity was more negative (maximal at Cz: −1.53 μV). The Task × Trial type × Group interaction suggested that HA exhibit greater activity on 'switch' ($M = 1.02$ μV) compared to 'repeat' ($M = -.37$ μV) prosaccade trials, $t(11) = 2.75$, $p = .02$, whereas LA exhibited comparable activity on both 'switch' ($M = .64$ μV) and 'repeat' ($M = .97$ μV) prosaccade trials, $t < 1$. There were no group-related differences between 'switch' and 'repeat' antisaccade trials.

3.5.7. Intermediate preparation interval (long CTI): 200–400 ms after instruction offset

A 2 × 2 × 2 × 6 mixed ANOVA on cortical activity with Group (LA, HA) as the between- and Task (anti, prosaccade), Trial type (switch, repeat), and Site (F3/F4, Fz, C3/C4, Cz, P3/P4, Pz) as the within-subject factors yielded a significant main effect of Site, $F(5,110) = 20.54$, $p < .001$, suggesting that activity at frontal and central sites was more negative (maximal at Cz: −1.84 μV) whereas activity at parietal sites was more positive (maximal at Pz: 1.67 μV). More importantly, the significant Site × Group, $F(5,110) = 3.69$, $p = .03$, interaction suggested that HA compared to LA individuals showed greater negativity at frontal (F3/F4 HA: −1.74 μV and LA: .7 μV, $t(22) = 2.38$, $p = .03$, Fz HA: −2.89 μV and LA: −.43 μV, $t(22) = 2.13$, $p = .04$) sites but there were no anxiety-related group difference at central and parietal sites, $ts < 1$. This replicates observations in medium CTI trials.

3.5.8. Late preparation interval (long CTI): 400–1500 ms after instruction offset

A 2 × 2 × 2 × 6 mixed ANOVA on cortical activity with Group as the between- and Task (anti, prosaccade), Trial type (switch, repeat), and Site (F3/F4, Fz, C3/C4, Cz, P3/P4, Pz) as the within-subject factors yielded a significant main effect of Task, $F(1,22) = 13.82$, $p < .01$, and Site, $F(5,110) = 13.93$, $p < .001$. The main effect of Task suggests that cortical activity was generally more negative on antisaccade ($M = -1.41$ μV) compared to prosaccade ($M = -.06$ μV) trials, while the main effect of Site indicated that activity was more negative at central and parietal sites (maximal at Cz: −2.48 μV) compared with frontal sites (maximal at F3/F4: .86 μV), mean diff = 3.34, $p < .001$. Importantly, there were no group differences during this late preparation period.

3.6. Summary (long CTI trials)

The key findings here were that, as with medium CTI trials, preparation-related ERP activity during the long CTI showed greater frontal CNV in HA versus LA individuals at 200–400 ms after instructional cue offset. On long CTI trials, however, this negativity gradually dissipated and reached baseline at target onset.

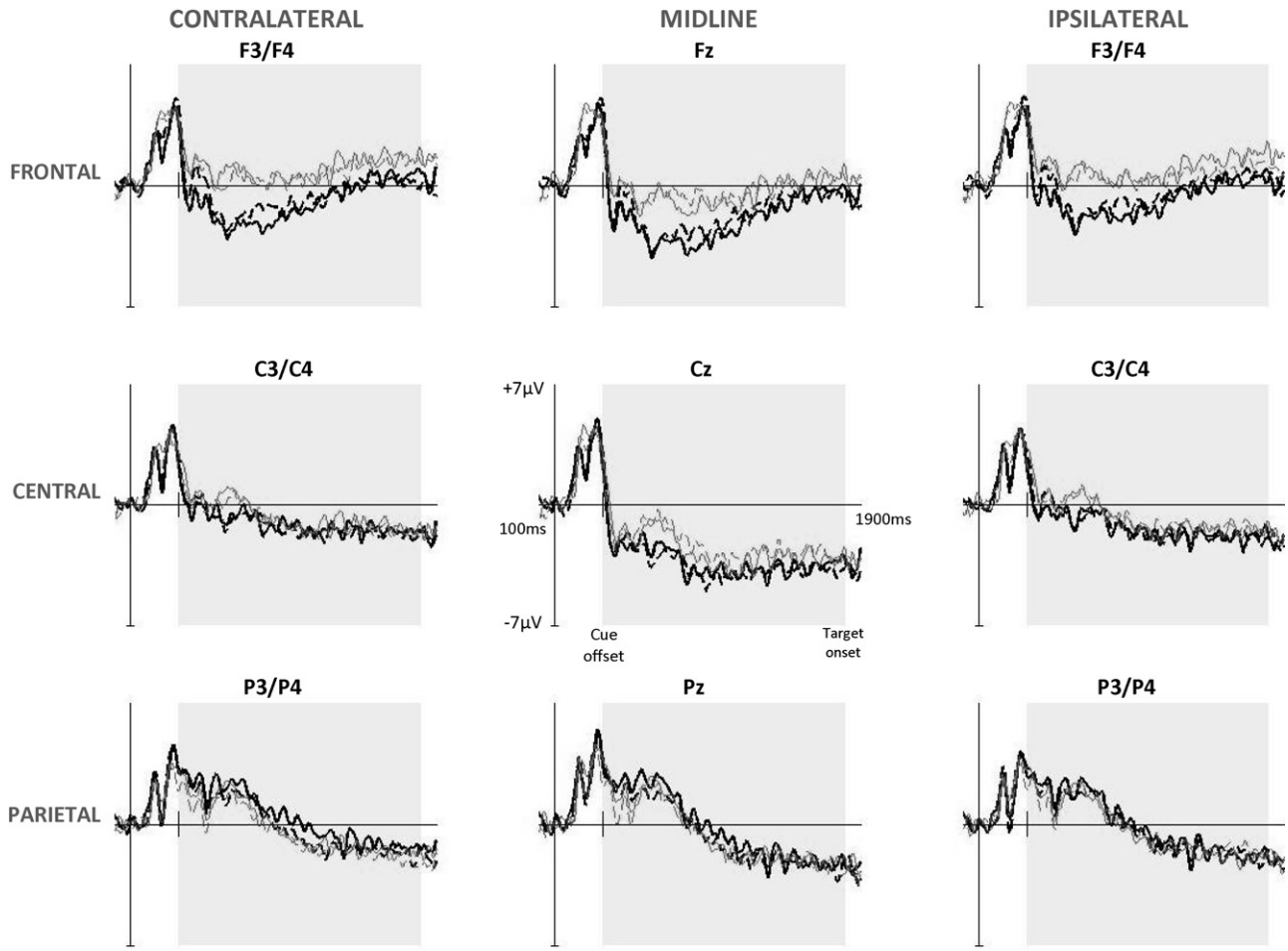

Fig. 6. Grand averages during long CTI at contralateral, ipsilateral and midline electrodes for HA (dark black lines) and LA (grey lines) on correct switch (solid lines) and repeat (dashed lines) antisaccade trials.

4. Discussion

This study aimed to evaluate the prediction from attentional control theory (Eysenck et al., 2007) that high-anxious individuals exert greater compensatory effort (and expend more processing resources) to maintain performance effectiveness comparable to that of low-anxious individuals. Our previous work has shown that anxiety is associated with impaired efficiency when central executive functions are involved (see Derakshan and Eysenck, 2009, for a review). In the current study, we assessed EEG activity and behavioural performance in a version of the mixed antisaccade task that manipulated the interval between the instructional cue and the target (short, medium, and long CTI). The findings corroborated our central predictions from ACT. Firstly, that high- versus low-anxious participants would exhibit greater cognitive effort as measured by frontal CNV activity (indicating increased allocation of cognitive resources) during medium and long CTI trials. Secondly, correct antisaccade latencies would be faster on medium and long compared with short CTI trials resulting in comparable switching performance between high- and low-anxious individuals. Our results supported these predictions.

EEG analysis found that the offset of instructional cue, which signalled the imminent need for a speeded response, induced a preparatory CNV potential at frontal cortical sites which was greater in high- compared with low-anxious individuals. This was observed on both medium and long CTI trials approximately 200–400 ms after instructional cue offset. Greater frontal CNV amplitudes in response to a warning signal (in this case the offset of instructional cue) have commonly been associated with increased allocation of cognitive resources to prepare cortical networks involved in responding (e.g. Bender et al., 2004; Cui et al., 2000; Jennings and van der Molen, 2005; Rösler et al., 1997). This well-established slow-wave potential is believed to originate from the PFC (e.g. Gómez et al., 2007; Rosahl and Knight, 1995) partly due to its frontal scalp distribution and partly because it occurs during processes that are believed to require PFC involvement (e.g. anticipation, planning and preparation). Our observations thus suggest that high-anxious participants exerted greater cognitive effort (and expended more resources) compared with the low-anxious in preparation for the task goal i.e. correct saccade performance. The behavioural data corroborated this interpretation. Specifically correct saccade latencies revealed that on medium and long CTI trials where target appeared after a lag (400 ms and 1500 ms respectively), everyone showed an improvement in correct saccade latencies compared with short CTI trials. There were no anxiety-related differences in saccade accuracy rates on these trials. These observations support the assumptions of attentional control theory (ACT; Eysenck et al., 2007). According to ACT, anxious individuals are motivated to avoid aversive outcomes that may result from poor task performance. Hence they should use compensatory strategies (such as enhanced effort or increased use of processing resources) that aim to prevent decrements in performance. However, this would come at a cost to processing efficiency, what is referred to

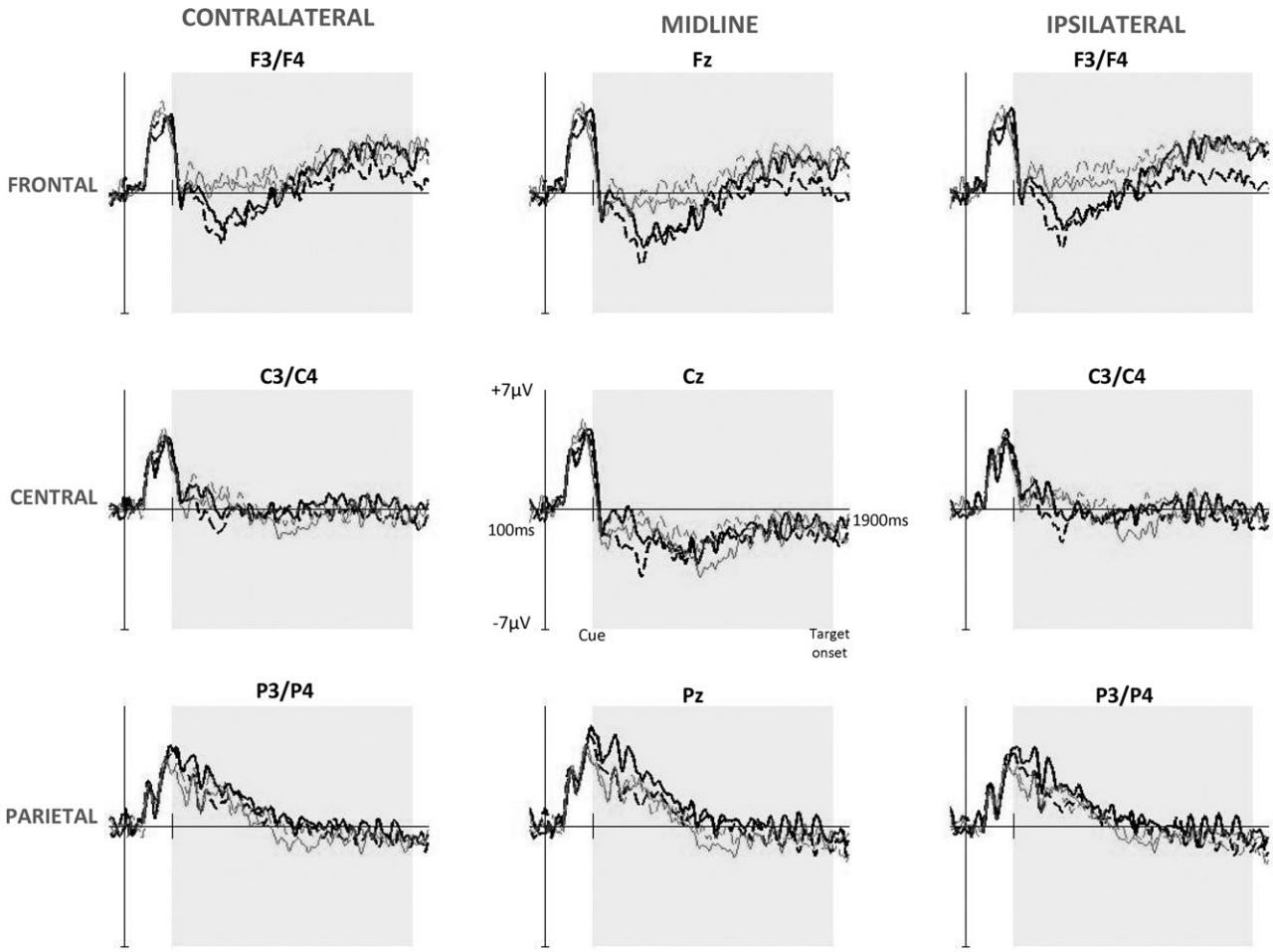

Fig. 7. Grand averages during long CTI at contralateral, ipsilateral and midline electrodes for HA (dark black lines) and LA (grey lines) on correct switch (solid lines) and repeat (dashed lines) prosaccade trials.

as the '*hidden cost*' of anxiety (Eysenck and Calvo, 1992; Eysenck et al., 2007).

Additionally, during the medium CTI trials greater frontal CNV activity in high-anxious individuals persisted until target onset which could indicate a state of sustained 'preparedness' (Jennings and van der Molen, 2005; Rösler et al., 1997). With long CTIs however, CNV activity dissipated gradually toward baseline such that at the time of target onset high- and low-anxious individuals exhibited comparable CNV activity at all recording sites. These observations are best interpreted with respect to behavioural performance following this activity. Correct saccade latencies revealed a marked improvement in all participants when target onset was preceded by a 400 ms preparation interval (medium CTI) compared with when target appeared immediately after instruction presentation (short CTI trials). As Fig. 2 showed, high-anxious individuals showed comparable saccade latencies to that of low-anxious individuals in both medium and long CTI trials, indicating that the introduction of the preparatory period resulted in an enhanced improvement in saccadic latencies for both but especially the high-anxious group. Interestingly, increasing the pre-target lag to 1500 ms (long CTI) did not lead to further latency improvements.

The behavioural data suggested that increasing CTI from 400 ms to 1500 ms was ineffective for low-anxious and possibly counterproductive for high-anxious individuals. Previous research highlights the notion of 'optimum preparedness' beyond which further preparation time does not improve performance (for a review, see Jennings and van der Molen, 2005). Our neural and behavioural data suggest that while anxious individuals were motivated to achieve the task goal – indicated by greater frontal CNV activity and faster latencies on medium CTI trials, they were less able to sustain a high level of attentional control during long cue-target intervals. This finding is in line with Fales et al. (2008) who associated enhanced anxiety with decreased 'sustained' executive control which was argued to be compensated for by increased 'transient' or reactive executive control. According to Fales et al. (2008) transient increases in cognitive control enables high-anxious individuals to maintain a level of performance that is comparable to that of the low-anxious group. The pattern of activity observed in the current study coupled with our behavioural results supports this argument.

Finally, analysis of accuracy rates showed that participants made more errors on short versus medium and long CTI trials, on antisaccade versus prosaccade trials and on 'switch' versus 'repeat' trials. However, there were no anxiety-related group differences in this measure of performance. These observations replicate our previous findings (e.g. Ansari et al., 2008; Derakshan et al., 2009; Ansari and Derakshan, 2010) and are in line with the view that anxious individuals are able to perform the task correctly (indicated by comparable accuracy rates) but they are less efficient at doing so (as indicated by the EEG data).

Our findings provide compelling support for the predictions of the attentional control theory (Eysenck et al., 2007) showing that anxious individuals use compensatory strategies such as increased cognitive effort to prevent performance decrements but that this comes at a cost to processing efficiency. This study is among the first to use a well-known neural index of resource allocation and cognitive effort to evaluate the relationship between anxiety and

processing efficiency (see also MacNamara and Hajcak, 2010). It will be interesting for future research to examine processing efficiency costs in relation to emotional material. Recently, MacNamara and Hajcak (2010) examined behavioural and ERP measures of attention allocation to aversive stimuli among individuals diagnosed with generalised anxiety disorder (GAD) and healthy controls. They found that GAD individuals exhibited diminished processing effectiveness (indicated by greater errors in response to aversive distracters) and impaired processing efficiency (indicated by greater attention-related neural activity) when aversive targets were paired with neutral distracters. The current study could be extended to include emotional material as targets and measure CNV related activity in relation to the inhibition of these targets. Indeed previous work from our lab shows that anxious individuals exhibit decreased efficiency when inhibiting threat related material as indicated by their longer antisaccade latencies from angry facial expressions of emotion (Derakshan et al., 2009).

The current study highlights a number of issues that require further investigation. For example, it is not clear which cognitive processes represent greater preparatory activity. It may indicate greater allocation of resources to processes such as task-rule activation/retrieval or working memory updating – processes that are known to be involved in antisaccade and task-switching performance (e.g. Mayr and Kliegl, 2000, 2003; Monsell, 2003; Rubinstein et al., 2001). Or high-anxious individuals find it hard to maintain attention on the task at hand and the onset of the instructional cue triggers a mechanism that helps to 'refocus' attention. This would be in line with the notion that anxious individuals are more vulnerable to distraction by task-irrelevant, self-focused thoughts and hence find it harder to remain focused over long periods of time. Support for the idea that CNV activity could be related to changes in conscious focus of attention comes from studies showing that CNV is enhanced when participants are specifically instructed (or given monetary reward) to focus on the imperative target. Evidence also comes from studies showing that *distraction* by engaging stimuli during preparation reduces CNV amplitudes (see Tecce, 1972, for a comprehensive review of this literature).

Another important question relates to the circumstances in which high-anxious individuals are motivated to exert compensatory effort. In the current study, it would have been fair to assume that more effort might be necessary for the (presumably) more difficult 'switch' trials compared with the 'repeat' trials. However, trial type did not influence CNV activity during the critical period of interest (i.e. 200–400 ms after cue offset) for any of the groups. This could be due to the fact that 'switch' and 'repeat' trials were randomly presented within an overall difficult mixed-task block encouraging participants to prepare to the same extent for *both* trial types. On the other hand it could be that high-anxious individuals generally exert more effort in their performance. The latter case seems more likely in the current experiment as low-anxious individuals also showed no differential CNV activity between 'switch' and 'repeat' trial types. It is theoretically important to determine whether or not there are specific circumstances where high-anxious individuals are more (or even less) inclined to use compensatory strategies to achieve the task goal.

Finally, our results may seem contradictory to the current debates that associate anxiety with impaired recruitment of prefrontal areas implicated in attentional control (see Bishop, 2009). It is important to note that the current study examined top down working memory functions in a design which motivated a preparatory response following a warning signal prior to target onset, whereas in Bishop (2009) spatial attention was assessed in a design that manipulated perceptual load functions and lower prefrontal activity was found immediately prior to target. Indeed recent observations show that anxiety is associated with lower ERP activity immediately before target onset in the antisaccade task (Ansari and Derakshan, in press). The current study assessed neural activity in a design that manipulated the preparatory period prior to the execution of the task goal to allow for a more detailed examination of the time course of processing efficiency in anxiety.

In conclusion this study was among the first to assess the proposed relationship between anxiety and cognitive effort by comparing CNV activity in high- and low-anxious individuals during performance of a demanding mixed antisaccade task. The results support the association between anxiety and increased effort and resource allocation to task performance (Eysenck and Calvo, 1992; Eysenck et al., 2007) indicating that when cognitive resources are available for effortful processing, high-anxious individuals can achieve performance that is comparable to the low-anxious. Future research should investigate the underlying processes involved in effortful processing and the circumstances that motivate greater effort in anxiety.

Acknowledgments

This work was supported by an Economic and Social Research Council (ESRC) PhD Studentship awarded to Tahereh L. Ansari and carried out under the supervision of Nazanin Derakshan. Many thanks to Max Owen for his help with data processing.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biopsycho.2010.12.013.

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